SGA 2: Application to the Fundamental Group

Bounded source-aligned working unit — 20 July 2026

Authority. Corrected French TeX lines 3433–3437; original printed pp. 116–117, physical source-PDF p. 101, and recomposed running p. 93. The `\pageoriginale` marker within line 3433 follows "two" in the phrase "two nonzero rings" and advances the original printed page. These locator systems are kept distinct. Blank line 3438 is excluded; the raw cursor is line 3438 and the next substantive cursor is line 3439 (Corollary 2.5). The connectivity criterion, direct images, global sections, hats, subscripts, and isomorphism were checked directly. The external English candidate is comparison-only.

Proof. Indeed, a locally ringed space $(X, \mathcal{O}_X)$ is connected if and only if $\Gamma(X, \mathcal{O}_X)$ is not a direct product of two nonzero rings. But, by $\mathrm{Lef}(X, Y)$, one has

$$\Gamma\big(U, r_*\mathcal{O}_R\big) \simeq \Gamma\big(\widehat{X}, \widehat{r}_*\mathcal{O}_{\widehat{R}}\big).$$